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Investor Presentation | Japan

Japan Summer School: Energy & Utilities

This presentation covers the fundamentals of Japan's Energy & Utilities industry, focusing on key topics including industry structure and market trends.

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ENERGY & UTILITIES

Japan

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In-Line

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Japan Summer School 2026



The Future of Energy in Japan 2026

[A] Contents

- (1) Japan's 7th Strategic Energy Plan (February 2025)
- (2) The 17 strategic sectors under the Takaichi administration
 - total public and private investment to exceed JPY 370 trillion (cumulative through 2040)
 - 12th strategic sector: Resources, Energy, Security & GX (Green Transformation)
 - 13th strategic sector: Fusion Energy (Nuclear Fusion Energy)
- (3) Power Demand Outlook — new construction of AI data centers and semiconductor fabrication plants
- (4) Power Generation Mix
- (5) Power Supply-Demand Balance Outlook
- (6) Nuclear Power Plants

Japanese Government's 7th Strategic Energy Plan: F3/41 Outlook

Outlook on energy supply and demand in F3/41

	F3/24 (Preliminary figures)	F3/41e
Energy self-sufficiency rate	15.2%	Approx. 30-40%
Electricity output	985.4bn kWh	Approx. 1.1-1.2trn kWh
Final energy consumption	300mn kl	Approx. 260-270mn kl
GHG emission reduction (vs F3/14)	22.9% (F3/23 actual)	73.0%

Outlook on breakdown of electricity output

	F3/24 (Preliminary figures)	F3/41e
Renewables	22.9%	Approx. 40-50%
Nuclear	8.5%	Approx. 20%
Thermal	68.6%	Approx. 30-40%

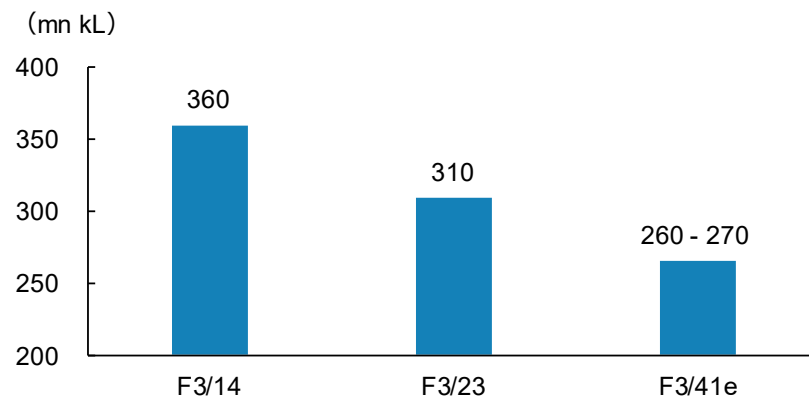
Outlook on breakdown of renewables in energy mix

	F3/24 (Preliminary figures)	F3/41e
Renewables	22.9%	Approx. 40-50%
Solar power	9.8%	Approx. 23-29%
Wind power	1.1%	Approx. 4-8%
Hydro	7.6%	Approx. 8-10%
Geothermal	0.3%	Approx. 1-2%
Biomass	4.1%	Approx. 5-6%

Source: METI, Morgan Stanley Research; e = METI outlook

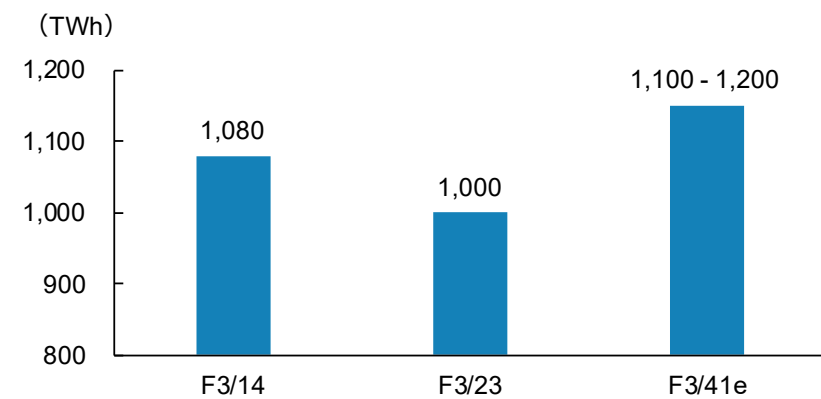
Japan: Outlook on Final Energy Consumption and Electricity Output for F3/41

Outlook on final energy consumption



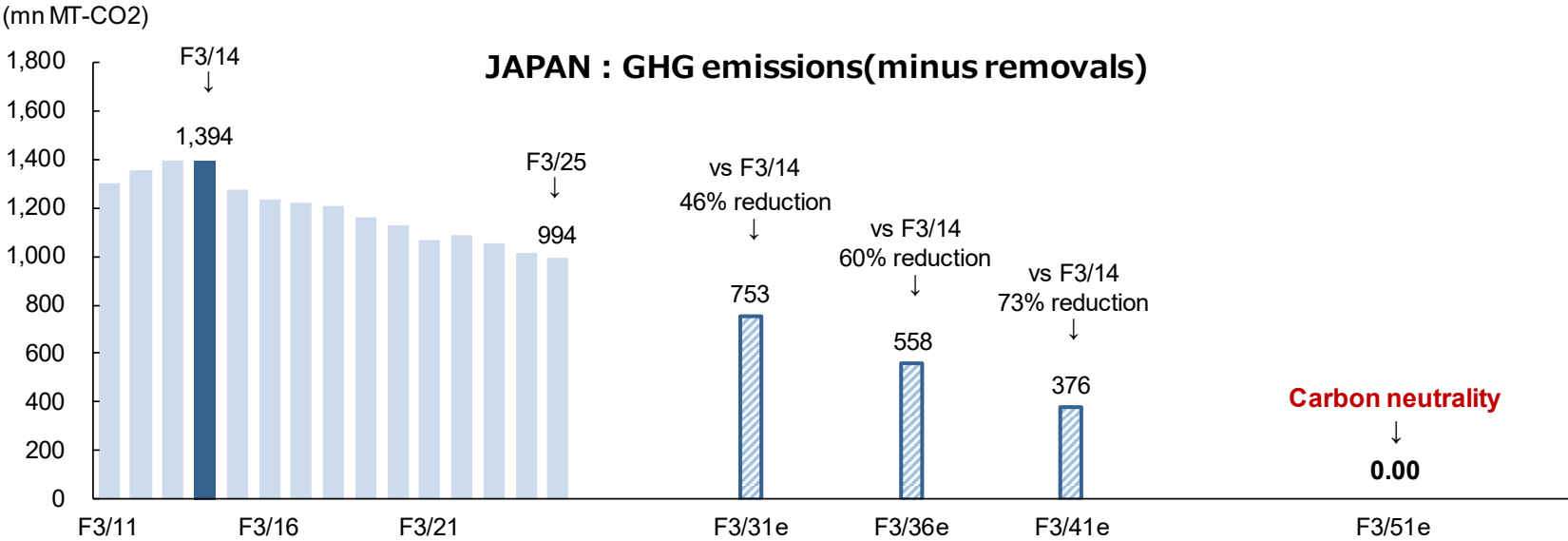
Source: METI, Morgan Stanley Research; e = METI outlook,
1PJ = 25,800 kL (crude oil equivalent)

Outlook on electricity output



Source: METI, Morgan Stanley Research; e = METI outlook

Japan: GHG Emissions



Source: MoE, Morgan Stanley Research Note: e=Morgan Stanley estimates

The 17 strategic sectors under the Takaichi administration — Total public and private investment to exceed JPY 370 trillion (cumulative through F2040)

The 17 Strategic Sectors under the Takaichi Administration

1	AI & Semiconductors
2	Digital & Cybersecurity
3	Information & Communications
4	Quantum
5	Defense Industry
6	Aviation & Space
7	Ocean/Maritime
8	Shipbuilding
9	Materials (Critical Minerals & Components)
10	Synthetic Biology & Biotechnology
11	Drug Discovery & Advanced Medicine
12	Resources, Energy, Security & GX
13	Fusion Energy
14	Disaster Prevention & National Resilience
15	Ports & Logistics
16	Food Tech
17	Content

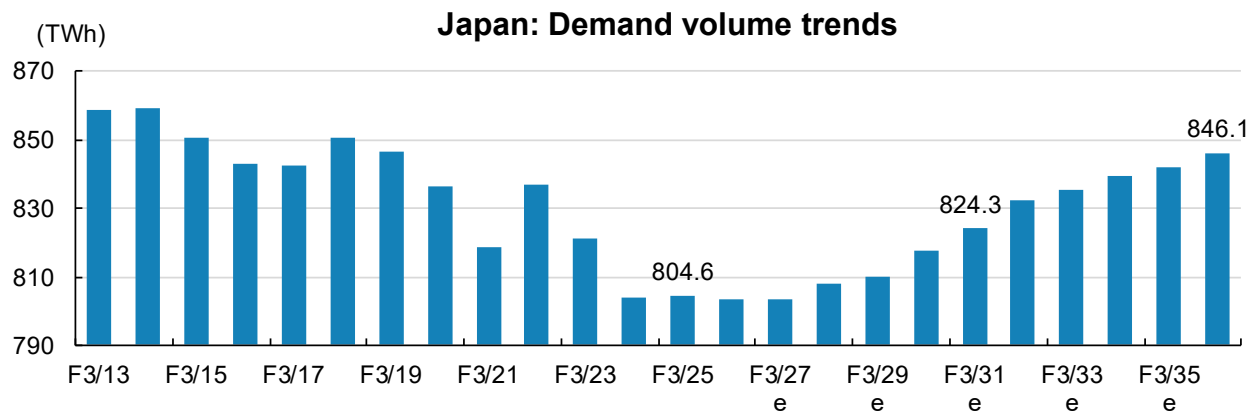
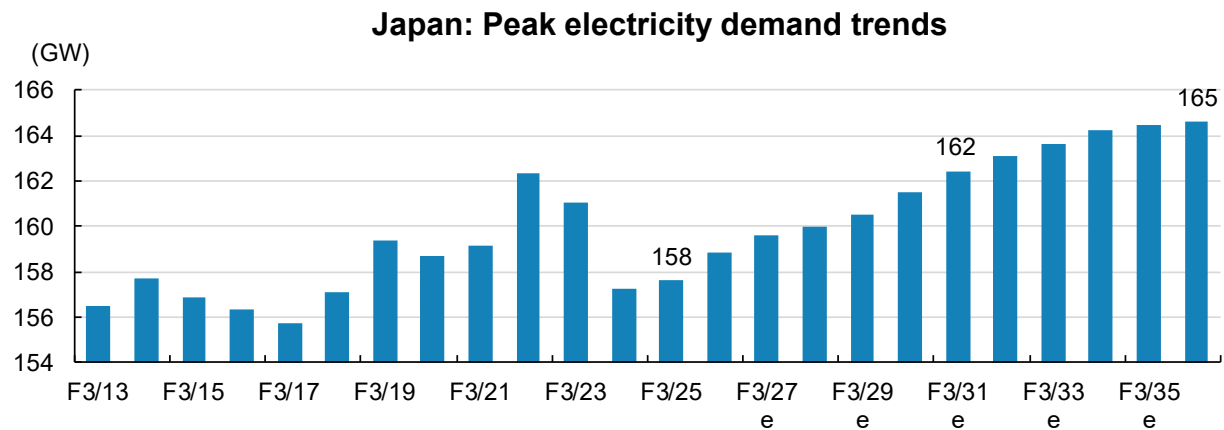
Source: Cabinet Secretariat, Morgan Stanley Research

Public and Private Investment relating to the Energy & Utilities Industry

		Investment (through F2040) (JPY trillion)
12	Resources, Energy, Security & GX	28.8
	a Next-Generation Solar Cells (e.g., Perovskite Solar Cells)	4.1
	b Hydrogen	6.2
	c Green Steel	4.2
	d Next-Generation Geothermal	1.0
	e Offshore Wind	5.1
	f Next-Generation Advanced Reactors	5.0
	g GX Chemicals	3.2
13	Fusion Energy	3.1
TOTAL(all 17 strategi sectors)		370

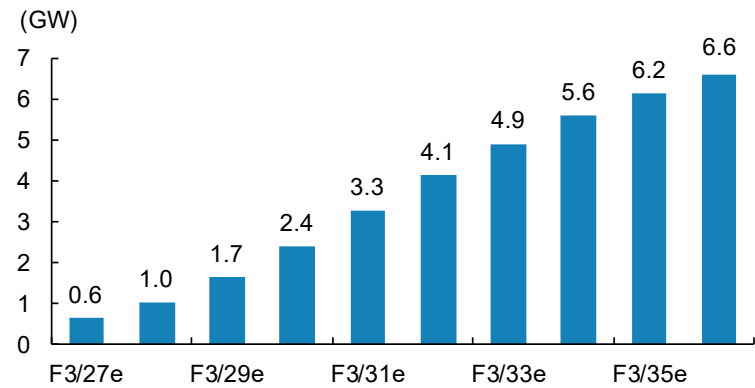
Source: Cabinet Secretariat, Morgan Stanley Research

Japan: Projected Electricity Demand (10-yr)



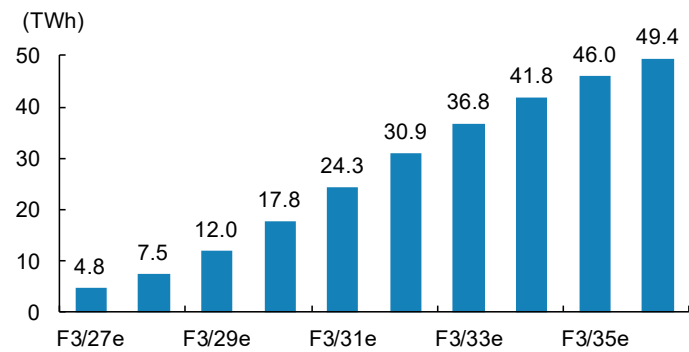
Japan: Projected Electricity Demand for Construction of New Data Centers

Projected rise in peak electricity demand from construction of new data centers (vs. F3/26)



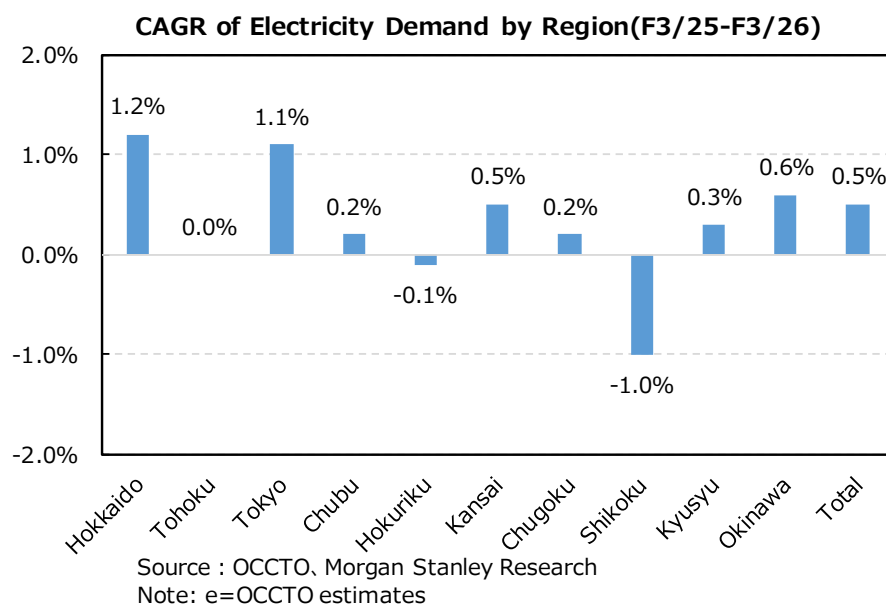
Source: OCCTO, Morgan Stanley Research; e = OCCTO estimates

Projected rise in demand volume from construction of new data centers (vs. F3/26)



Source: OCCTO, Morgan Stanley Research; e = OCCTO estimates

Outlook of Electricity Demand by Region



Outlook of Electricity Demand by Region

	(TWh)		
	F3/26e	F3/36e	CAGR (F3/25-F3/36)
Hokkaido	27.8	31.2	1.2%
Tohoku	74.0	73.7	0.0%
Tokyo	257.6	287.1	1.1%
Chubu	121.8	124.6	0.2%
Hokuriku	25.7	25.3	-0.1%
Kansai	130.9	137.0	0.5%
Chugoku	53.1	54.2	0.2%
Shikoku	23.7	21.4	-1.0%
Kyusyu	80.9	83.2	0.3%
Okinawa	7.9	8.4	0.6%
Total	803.4	846.1	0.5%

Source : OCCTO, Morgan Stanley Research
Note : e=OCCTO estimates

Japan: Power Generation by Energy Source

Power generation by energy source

(TWh)	F3/14		F3/23	F3/24	F3/25			
		% of total				YoY	vs F3/14	% of total
Total	1,084.5	100%	1,001.7	987.6	991.1	0%	-9%	100%
Coal	357.1	33%	304.2	281.2	278.5	-1%	-22%	28%
Natural gas	443.5	41%	338.5	323.2	319.4	-1%	-28%	32%
Oil, etc.	156.7	14%	84.1	73.1	71.1	-3%	-55%	7%
Nuclear	9.3	1%	56.1	84.1	93.5	11%	905%	9%
Renewable energy	117.9	11%	218.7	226.1	228.6	1%	94%	23%
Hydroelectric power	79.4	7%	76.7	74.9	73.5	-2%	-7%	7%
Solar power	12.9	1%	92.6	96.5	98.1	2%	660%	10%
Wind power	5.2	0%	9.3	10.5	11.7	11%	125%	1%
Geothermal power	2.6	0%	3.0	34.0	3.9	-89%	50%	0%
Biomass power	17.8	2%	37.2	40.8	41.4	1%	133%	4%

Source: MoE, Morgan Stanley Research

Japan: Supply & Demand Balance of Electricity

Outlook of Supply and Demand balance of Electricity (Aug 2027)

(MW)	Supply capacity (a)	Peak demand (b)	Surplus capacity (c) = (a)-(b)	Surplus capacity ratio (c) / (a)
East area	85,610	72,510	13,100	15%
Hokkaido	5,220	4,220	1,000	19%
Tohoku	15,820	13,020	2,800	18%
Tokyo	64,580	55,270	9,310	14%
Central and west area	101,050	90,590	10,460	10%
Chubu	27,690	23,700	3,990	14%
Hokuriku	5,460	4,670	790	14%
Kansai	31,470	26,930	4,540	14%
Chugoku	11,710	10,020	1,690	14%
Shikoku	6,140	4,640	1,500	24%
Kyushu	18,590	15,910	2,680	14%
Total of 9 areas	186,670	158,380	28,290	15%

Source: OCCTO, Morgan Stanley Research; Note: OCCTO outlook

Japan: Power Supply Capacity, Status of Nuclear Power Plants

Japan : Supply Capacity of Electricity

(MW)	F3/26
Thermal Power	142,030
Coal	51,380
LNG	78,450
Oil etc	12,200
Nuclear Power	33,080
Renewable power etc	146,230
Hydropower	22,140
Pumped-Storage Power	27,340
Wind	6,840
Solar	80,190
Geothermal	510
Biomass	7,180
Waste-to-Energy	1,350
Battery Energy Storage	660
Hydrogen	0
Ammonia	0
Others	2,310
Total	326,350

Source : OCCTO、Morgan Stanley Research

Note : e=OCCTO outlook

Status of Nuclear Power Plants

2026/6/26	Power capacity total		Number of NPP	
	(MW)	Already restarted (MW)	(units)	Already restarted (units)
East Area	14,132	2,181	14	2
West Area	18,951	12,428	19	13
Total	33,083	14,609	33	15

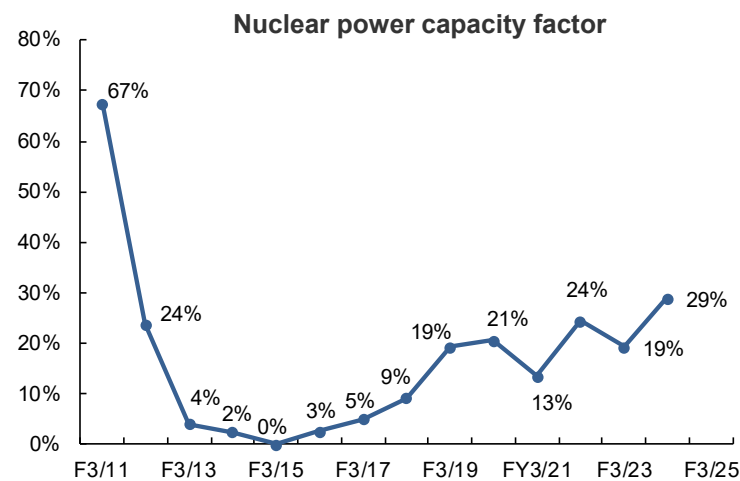
Source: Morgan Stanley Research

Domestic Nuclear Power Plants

Company	Nuclear power plant	Power generation capacity (MW)	Prefecture	Start of operation	Type of reactor
9501 TEPCO HD	Kashiwazaki Kariwa No.1	1,100	Niigata	Sep 1985	BWR
	Kashiwazaki Kariwa No.2	1,100	Niigata	Sep 1990	BWR
	Kashiwazaki Kariwa No.3	1,100	Niigata	Aug 1993	BWR
	Kashiwazaki Kariwa No.4	1,100	Niigata	Aug 1994	BWR
	Kashiwazaki Kariwa No.5	1,100	Niigata	Apr 1990	BWR
	Kashiwazaki Kariwa No.6	1,356	Niigata	Nov 1996	ABWR
	Kashiwazaki Kariwa No.7	1,356	Niigata	Jul 1997	ABWR
9502 Chubu EP	Hamaoka No.3	1,100	Shizuoka	Aug 1987	BWR
	Hamaoka No.4	1,137	Shizuoka	Sep 1993	BWR
	Hamaoka No.5	1,380	Shizuoka	Jan 2005	ABWR
9503 Kansai EP	Mihama No.3	826	Fukui	Dec 1976	PWR
	Ohi No.3	1,180	Fukui	Dec 1991	PWR
	Ohi No.4	1,180	Fukui	Feb 1993	PWR
	Takahama No.1	826	Fukui	Nov 1974	PWR
	Takahama No.2	826	Fukui	Nov 1975	PWR
	Takahama No.3	870	Fukui	Jan 1985	PWR
	Takahama No.4	870	Fukui	Jun 1985	PWR
9504 Chugoku EP	Shimane No.2	820	Shimane	Feb 1989	BWR
9505 Hokuriku EP	Shika No.1	540	Ishikawa	Jul 1993	BWR
	Shika No.2	1,206	Ishikawa	Mar 2006	ABWR
9506 Tohoku EP	Higashidori No.1	1,100	Aomori	Dec 2005	BWR
	Onagawa No.2	825	Miyagi	Jul 1995	BWR
	Onagawa No.3	825	Miyagi	Jan 2002	BWR
9507 Shikoku EP	Ikata No.3	890	Ehime	Dec 1994	PWR

Source: JAIF, Morgan Stanley Research

Company	Nuclear power plant	Power generation capacity (MW)	Prefecture	Start of operation	Type of reactor
9508 Kyushu EP	Genkai No.3	1,180	Saga	Mar 1994	PWR
	Genkai No.4	1,180	Saga	Jul 1997	PWR
	Sendai No.1	890	Kagoshima	Jul 1984	PWR
	Sendai No.2	890	Kagoshima	Nov 1985	PWR
9509 Hokkaido EP	Tomari No.1	579	Hokkaido	Jun 1989	PWR
	Tomari No.2	579	Hokkaido	Apr 1991	PWR
	Tomari No.3	912	Hokkaido	Dec 2009	PWR
Unlisted Japan Atomic Power	Tokai No.2	1,100	Ibaraki	Nov 1978	BWR
	Tsuruga No.2	1,160	Fukui	Feb 1987	PWR



Source: MoE data, Morgan Stanley Research

Japan: Key Businesses in the Energy/Utilities Industry

[A] Electricity business

- (1) Electricity businesses are broadly divided into [a] power generation, [b] transmission/ distribution, and [c] retail businesses
- (2) The power generation business is not regulated by the Japanese government. The power transmission/distribution business is regulated by the Japanese government.
- (3) The retail business was fully deregulated in April 2016. However, part of the retail business remains regulated by the Japanese government (transitional measures)
- (4) Examples companies: Chubu Electric Power (9502.T), Kansai Electric Power (9503.T), Japan Petroleum Exploration (1662.T)

[B] City gas business

- (1) The city gas business is divided into [a] operation of LNG base, [b] LNG import, [c] distribution of city gas via pipes, and [d] retail
- (2) The city gas distribution business is a regulated by the Japanese government
- (3) Was fully deregulated in April 2017. Currently, the retail field is not regulated by the Japanese government.
- (4) Example companies: Tokyo Gas (9531.T), Osaka Gas (9532.T)

[C] Crude oil and natural gas upstream business (E&P business)

- (1) Overview of the upstream business: Produce crude oil and natural gas in oil and gas fields and sell these.
- (2) Example companies: INPEX (1605.T), Japan Petroleum Exploration (1662.T)

[D] Refining business of petroleum products

- (1) Overview of the petroleum product refinery business: Produce petroleum products from crude oil at refineries and wholesale them.
- (2) Example companies: Idemitsu Kosan (5019.T), ENEOS Holdings (5020.T), Cosmo Energy Holdings (5021.T)

[E] LP gas sales business

- (1) Overview of the LP gas sales business: Procure LP gas (propane, etc.) externally and sell them (wholesale or retail)
- (2) Example company: Iwatani (8088.T)

Map of Japan



Source: GSI (The Geospatial Information Authority of Japan), Morgan Stanley Research

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: Energy & Utilities

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/21/2026)
Reiji Ogino		
Chubu Electric Power (9502.T)	E (03/01/2024)	¥2,952
Chugoku Electric Power (9504.T)	O (03/01/2024)	¥941
Cosmo Energy Holdings (5021.T)	O (12/03/2024)	¥3,930
Electric Power Development (9513.T)	U (03/01/2024)	¥3,765

ENEOS Holdings (5020.T)	++	¥1,301
Hokkaido Electric Power (9509.T)	O (12/17/2024)	¥1,018
Hokuriku Electric Power (9505.T)	E (03/01/2024)	¥938
Idemitsu Kosan (5019.T)	E (03/01/2024)	¥1,293
INPEX (1605.T)	O (07/12/2021)	¥3,580
Iwatani (8088.T)	E (06/06/2025)	¥1,975
Japan Petroleum Exploration (1662.T)	E (12/04/2017)	¥1,718
Kansai Electric Power (9503.T)	E (12/11/2024)	¥2,356
Kyushu Electric Power (9508.T)	O (07/02/2026)	¥1,782
Osaka Gas (9532.T)	E (11/18/2024)	¥5,436
Shikoku Electric Power (9507.T)	E (03/01/2024)	¥1,520
Toho Gas (9533.T)	U (03/01/2024)	¥1,204
Tohoku Electric Power (9506.T)	E (07/02/2026)	¥1,146
TOKYO GAS (9531.T)	E (03/01/2024)	¥6,127

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* Historical prices are not split adjusted.